GATE ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 03



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Recap of Previous Lecture



Topic Clocks

Time → Angle

Angle → Time

Topics to be Covered



Topic-1 More on Clocks

Topic-2 Understanding concept of Average

Topic-3 Approaching questions with and without formula

Questionnaire:

$6:21\frac{9}{11}$

$6:43\frac{7}{11}$

Assignment

#Q. In between 6 0' clock and 7 0' clock at what time the hands of clock form 60°?

$$\frac{120^\circ}{5.5} = \frac{240}{11} = 21\frac{9}{11}$$

$$\frac{240}{5.5} = \frac{480}{11} = 43\frac{7}{11}$$



Questionnaire:

360 − 100

1 : $23\frac{7}{11}$

1 : $52\frac{8}{11}$

Assignment

#Q. In between 1 O' clock and 2 O' clock at what time the hands of clock form

100°? 260°

$\frac{130^\circ}{5.5} = \frac{260}{11} = 23\frac{7}{11}$



$\frac{290}{5.5} = \frac{580}{11} = 52\frac{8}{11}$

Questionnaire:

1:52 $\frac{8}{11}$

1 – 2 @ 100°

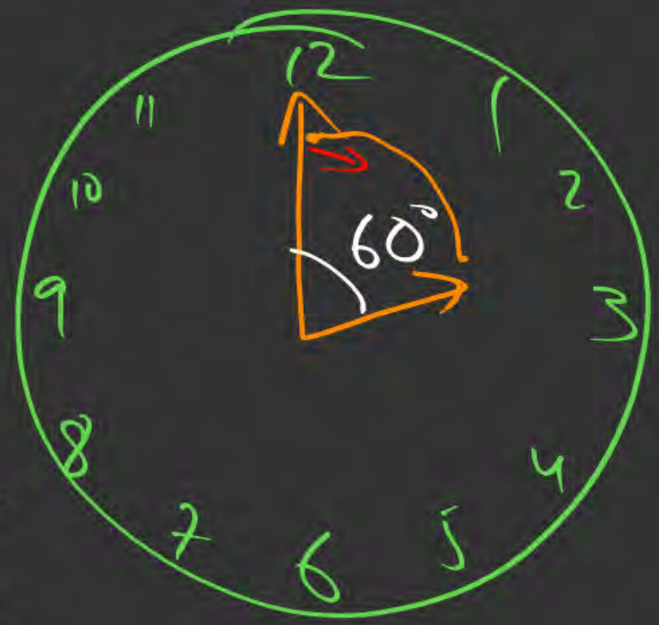
$\frac{40}{5.5} = \frac{80}{11}$

(–) = $7\frac{3}{11}$ min



Q. 2-3 @ 90° 270° $2:27\frac{3}{11}$

$$\frac{150^\circ}{5.5} = \frac{300}{11} = 27\frac{3}{11}$$



$$\frac{330}{5.5} = \frac{660}{11} = 60$$

Third Pattern:

Gain OR Lose

Questionnaire:

#Q. A Clock which gains 5 minutes in every one hour was set correct at 5am. What would be the time shown by that clock at 1pm the same day?

1:40 pm

Questionnaire:

120 min
OR
2hrs

#Q. A clock which looses 10 minutes in every one hour was set correct at 4am, what would be the time shown by that clock at 4pm the same day?

- 2hrs

CHAIN RULE

2pm

AVERAGE

→ A. Mean

Sum / No.

Normal Average → Mean

Middle Average → Median

fav Average → Mode

Inf^n ×

2, 3, 9, 0, 7

① $A = \frac{2+3+9+7}{4}$

$A.M. = \frac{2+3+9+0+7}{5}$

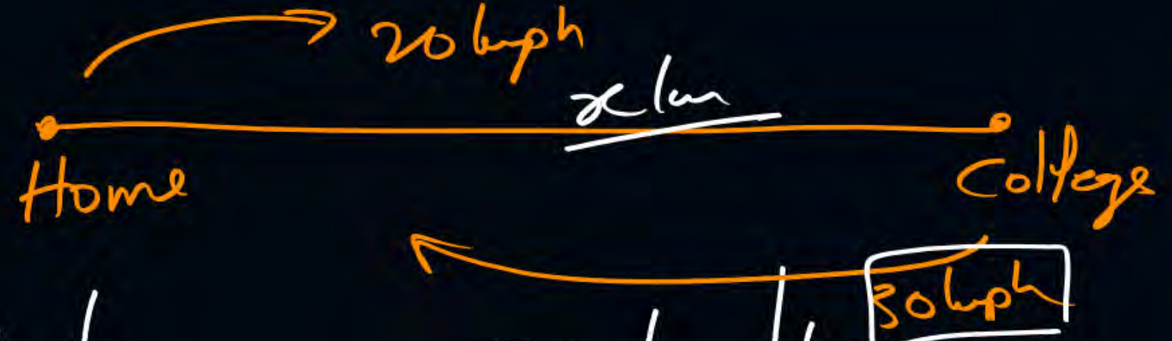
②

AVERAGE

$D = S \times T$

$S = \frac{D}{T}$

$T = \frac{D}{S}$



20 km/hr

30 km/hr

Average Speed $= 24$ km/hr

$$\frac{x}{20} + \frac{x}{30} = \frac{2x}{A.S.}$$

$$\Rightarrow \frac{5x}{60} = \frac{2x}{A.S.}$$

$$\Rightarrow A.S. = 24 \text{ km/hr}$$

Mean = $\frac{\text{Sum of obs.}}{\text{No. of obs.}}$

15

15

Equal Distribution



[MCQ]

#Q. What would be the average of: 1, 2, 3, 4, 5, ... 49, 50.

$$A = \frac{\text{Sum}}{\text{No.}}$$

2, 4, 6, 8, 10 → 6

2, 4, 6, 8, 10, 12 → 7

1275

$$\Sigma n = \frac{n(n+1)}{2}$$

$$\frac{n+1}{2}$$

$$x = 1+2+3 \ldots\ldots 49+50$$

$$x = 50+49+48 \ldots\ldots 2+1$$

$$2x = 50 \times 51$$

$$x = \frac{50 \times 51}{2} = 1275$$

Average of Even & Odd

Natural $\frac{n+1}{2}$

Even no $\Rightarrow n+1$

Odd $\Rightarrow n$

1, 3, 5, 7, 9 $\Rightarrow$ 5

1, 3, 5 $\Rightarrow$ 3

Sum of Even & Odd

$$A = \frac{Sum}{No.}$$

$$A \times No. = Sum$$

Starting

Sum of Even no $\Rightarrow (n+1) \times n = \underline{n(n+1)}$

Sum of odd no $\Rightarrow n \times n = \boxed{n^2}$

Sum / No.

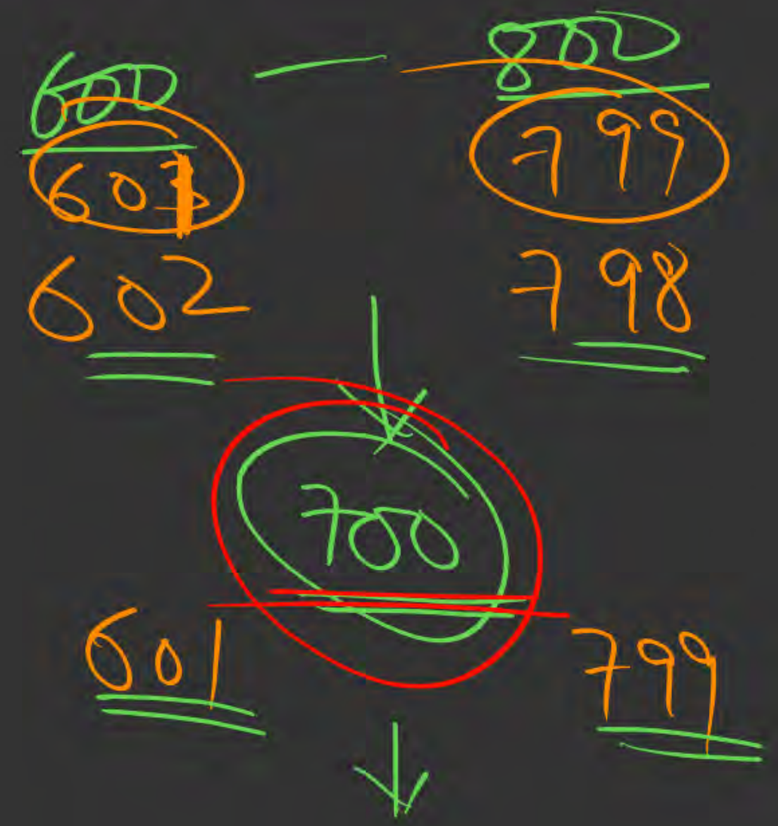
501 – 700

Even no between

(N) 502 – 699

(E) 502 – 698

(O) 503 – 699



[MCQ]

#Q. What is the average of first five multiples of 12?

A 42

B 40

C 36

D 48

12, 24, 36, 48, 60

[MCQ]

#Q. A class with 20 students has the average age as 14 years. When the teacher is included in the group, the average becomes 16 years. What would be the age of that teacher?

$A = \frac{Sum}{No.}$

$A \times No. = Sum$

$14 \times 20 = 280$

$16 \times 21 = 336$

$336 - 280 = 56$

$+\ 14$

42

56

[MCQ]

#Q. A team of 10 employees has the average age as 20 years. If team leader is excluded from the group the average of remaining 9 employees decreases by two months. Find the age of the team leader.

$19\frac{5}{6} \times 9$

$\Rightarrow \frac{119}{6} \times \frac{3}{2}$

$\Rightarrow = \frac{357}{2} = 178.5$

$A \times No. = Sum$

$20 \times 10 = 200$

$19\frac{10}{12} \times 9 = 178.5$

21.5

[MCQ]

#Q. In a school with 15 teachers, the average monthly salary is ₹4500. When three teachers left the school, the average monthly salary decreased by ₹600. Find the average monthly salary of three teachers who left the school.

Average × No = Sum

15 × 4500 = 67,500

12 × 3900 = 46,800

20,700

6900

20700 / 3

[MCQ]

#Q. The average score of a class of 40 students is 52. What will be the average score of the rest of the students if the average score of 10 of the students is 61.

Assignment

A 47

B 49

C 50

D 48

[MCQ]

#Q. The average weight of a school of 40 teachers is 80 kg. If, however, the weight of the principle be included, the average decreases by 1 kg. What is the weight of the principal?

Assignment

A 49

B 109

C 39

D 29

[MCQ]

#Q. The average age of Abhijeet and Daya is 20 years. Their average age 5 years hence will be

Assignment

A 20

B 30

C 25

D 22



2 mins Summary

Topic Averages

Average

Meaning

Formula

Without Formula

THANK - YOU